

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specifications for Bulk Items (Instrumentation)
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Specifications for Bulk Items (Instrumentation)

R02	23.10.22	Re-Issued for Review	K.Panchal	P.Roy	A.Duru	
R01	18.07.22	Issued for Review	K.Panchal	P.Roy	A.Duru	
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Revision History		
Rev No.	Date	Description of Revision
R01	18/07/2022	Issued for Review
R02	23/10/2022	Re-Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved “As-Built” documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the “As-Built” document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-024

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project’s FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, GAS, PLANT, ROW, PTS

REFERENCES

Document No.	Description

SPECIFICATIONS FOR BULK ITEMS (INSTRUMENTATION)

SL NO	ITEM SHORT DESCRIPTION	SPECIFICATION	BBOU	ELEP	AKAB	Total	UNITS
1	Junction Box						
1.1	Analog Junction Box for PLC (NIS) Non ESD	Analog Junctions boxes shall have 24 terminals suitable for 6 Pair X 1.5 mm2 cable main cable (3/4" NPT) and 1P X 1.5 mm2 (1/2" NPT) branch cable. 20% spare terminal shall also be provided. Exd, IP-65 for Zone1, IIA/IIB, T4. The certification shall be IEC/ATEX/FM/UL. MOC : SS316	2	4	2	6	Nos.
1.2	Analog Junction Box for PLC (NIS) ESD and FGS	Analog Junctions boxes shall have 24 terminals suitable for 6 Pair X 1.5 mm2 cable main cable (3/4" NPT) and 1P X 1.5 mm2 (1/2" NPT) branch cable or 1T X 1.5 mm2. 20% spare terminal shall also be provided. Exd, IP-65 for Zone1, IIA/IIB, T4. The certification shall be IEC/ATEX/FM/UL.MOC : SS316	4	5	2	8	Nos.
1.3	Digital Junction Box for PLC (NIS) Solenoid	Analog Junctions boxes shall have 24 terminals suitable for 6 Pair X 1.5 mm2 cable main cable (3/4" NPT) and 1P X 1.5 mm2 (1/2" NPT) branch cable. 20% spare terminal shall also be provided. Exd, IP-65 for Zone1, IIA/IIB, T4. The certification shall be IEC/ATEX/FM/UL.MOC : SS316	3	3	1	5	Nos.
1.4	Canopy for Junction Box	Three sided Canopy for Junction Box MS/GRP	9	12	5	19	Nos.
2	Cable						
2.1	1P X 1.5mm2 for Non ESD Outer sheet Black	Twisted, seven (7) stranded 1.5 sq.mm, annealed tinned copper conductors, XLPE insulated, with 0.5 mm2 annealed tinned copper drain wire, galvanised wire armoured, overall PVC sheathed, Fire Retardant, IEC-60331/60332	300	500	200	700	Mtrs
2.2	1P X 1.5mm2 for ESD/FGS Outer sheet Red	Twisted, seven (7) stranded 1.5 sq.mm, annealed tinned copper conductors, XLPE insulated, with 0.5 mm2 annealed tinned copper drain wire, galvanised wire armoured, overall PVC sheathed, Fire Retardant, IEC-60331/60332	800	900	800	2400	Mtrs

SPECIFICATIONS FOR BULK ITEMS (INSTRUMENTATION)

SL NO	ITEM SHORT DESCRIPTION	SPECIFICATION	BBOU	ELEP	AKAB	Total	UNITS
2.3	1P X 1.5mm2 for Solenoid Valve Gray	Twisted, seven (7) stranded 1.5 sq.mm, annealed tinned copper conductors, XLPE insulated, with 0.5 mm2 annealed tinned copper drain wire, galvanised wire armoured, overall PVC sheathed, Fire Retardant, IEC-60331/60332	300	400	200	700	Mtrs
2.4	1T X 1.5mm2 for ESD/FGS Outer sheet Red	Twisted, seven (7) stranded 1.5 sq.mm, annealed tinned copper conductors, XLPE insulated, with 0.5 mm2 annealed tinned copper drain wire, galvanised wire armoured, overall PVC sheathed, Fire Retardant, IEC-60331/60332	200	300	100	400	Mtrs
2.5	6P X 1.5mm2 for Non ESD Outer sheet Black	Multipair, twisted, seven (7) stranded 1.5 sq.mm, annealed tinned copper conductors, XLPE insulated, overall screen and with 0.5 mm2 annealed tinned copper drain wire, galvanised wire armoured, overall PVC sheathed. All the multi pair/multi triad cables shall be have both individual pair screen and overall shield for both digital and analog signal cables.Fire Retardant, IEC-60331/60332	100	200	200	500	Mtrs
2.6	6P X 1.5mm2 for ESD/FGS Outer sheet Red	Multipair, twisted, seven (7) stranded 1.5 sq.mm, annealed tinned copper conductors, XLPE insulated, overall screen and with 0.5 mm2 annealed tinned copper drain wire, galvanised wire armoured, overall PVC sheathed. All the multi pair/multi triad cables shall be have both individual pair screen and overall shield for both digital and analog signal cables.Fire Retardant, IEC-60331/60332	200	250	200	600	Mtrs
2.7	6P X 1.5mm2 for Solenoid Valve Gray	Multipair, twisted, seven (7) stranded 1.5 sq.mm, annealed tinned copper conductors, XLPE insulated, overall screen and with 0.5 mm2 annealed tinned copper drain wire, galvanised wire armoured, overall PVC sheathed. All the multi pair/multi triad cables shall be have both individual pair screen and overall shield for both digital and analog signal cables.Fire Retardant, IEC-60331/60332	150	150	100	350	Mtrs
2.8	Earthing Cable, Wire 6mm2 Cu Green FRLS, PVC	Wire Flexible Bare Annealed Copper 6mm2 (84/0.3); Sheating - FR PVC Colour-Green. IEC-60331	500	500	250	1000	Mtrs

SPECIFICATIONS FOR BULK ITEMS (INSTRUMENTATION)

SL NO	ITEM SHORT DESCRIPTION	SPECIFICATION	BBOU	ELEP	AKAB	Total	UNITS
3	Cable Gland and Plug						
3.1	Cable Gland 1/2" NPT	Double Compression with Lock Nut with PVC Hood. Exd, IP-65 for Zone1, IIA/IIB, T4. The certification shall be IEC/ATEX/FM/UL.MOC : SS316	50	70	24	98	Nos.
3.2	Cable Gland 3/4" NPT	Double Compression with Lock Nut with PVC Hood. Exd, IP-65 for Zone1, IIA/IIB, T4. The certification shall be IEC/ATEX/FM/UL.MOC : SS316	25	35	15	55	Nos.
3.3	Cable Gland 1" NPT	Double Compression with Lock Nut with PVC Hood. Exd, IP-65 for Zone1, IIA/IIB, T4. The certification shall be IEC/ATEX/FM/UL.MOC : SS316	10	15	10	30	Nos.
3.4	Cable Gland 1/2" ET	Single Compression with Lock Nut . Weather Proof IP-42. MOC : SS316	20	35	24	68	Nos.
3.5	Cable Gland 1" ET	Single Compression with Lock Nut . Weather Proof IP-42. MOC : SS316	10	15	10	30	Nos.
3.6	Cable Gland 3/4" ET	Single Compression with Lock Nut . Weather Proof IP-42. MOC : SS316	25	35	15	55	Nos.
3.7	Adaptor 1.5" NPT X 3/4" NPT	Explosion Proof. IP-65 for Zone1, IIA/IIB, T4. The certification shall be IEC/ATEX/FM/UL.MOC : SS316	20	25	10	40	Nos.
3.8	Plug 1" NPT	Hexagonal Plug ,Explosion Proof, IP-65 for Zone1, IIA/IIB, T4. The certification shall be IEC/ATEX/FM/UL.MOC : SS316	20	25	10	40	Nos.
3.9	Plug 3/4" NPT	Hexagonal Plug ,Explosion Proof, IP-65 for Zone1, IIA/IIB, T4. The certification shall be IEC/ATEX/FM/UL.MOC : SS316	20	25	10	40	Nos.
3.10	Plug 1/2" NPT	Hexagonal Plug ,Explosion Proof, IP-65 for Zone1, IIA/IIB, T4. The certification shall be IEC/ATEX/FM/UL.MOC : SS316	20	25	10	40	Nos.
4	Cable Ferrule						

SPECIFICATIONS FOR BULK ITEMS (INSTRUMENTATION)

SL NO	ITEM SHORT DESCRIPTION	SPECIFICATION	BBOU	ELEP	AKAB	Total	UNITS
4.1	Cable Ferrule for 1P X 1.5 mm2	SS Cable Ferrule,tentative diameter 15 mm	100	150	50	200	Nos.
4.2	Cable Ferrule for 6P X 1.5 mm2	SS Cable Ferrule,diameter 22 mm	100	150	50	200	Nos.
4.3	Cable Ferrule for 1P X 1.5 mm2	SS Cable Ferrule,diameter 15 mm	100	150	50	200	Nos.
5	Instrument Tag Plate						
5.1	Instrument Tag Plate	SS tag Plate of 50 mm X 200 mm size , 2mm thick with 0.5 mm hole, Letter Height and Width 4 mm X 4 mm.	40	50	20	80	Nos.
5.2	Cable Tag for Instrument Cable	SS Cable Tags 10 m X 90mm , 1 Line 30 Characters	40	50	20	80	Nos.
5.3	Tags for Valves	SS Valve Tags 60 mm X 15 mm , 1 Line 12 Character	4	6	4	12	Nos.
6	Cable Tie						
6.1	Cable Tie 150x2.5 Nylon 6.6	Cable Tie 150mm Long x 2.5mm Wide; MOC-Nylon 6.6	40	50	20	80	Nos
7	Bus Bar						
7.1	BusBar Supprt 30xM6 Con DMC	Conical Busbar Support Female/Female 30mm H x M6; MOC- Hi Dielectric DMC Polysterin	8	10	6	20	Nos
7.2	BusBar 25Wx5Tx100 CuTi	Bus Bar 25wide x 5Thick xCTC 100mm; MOC-Copper Tin Plated.	8	10	6	20	Nos
8	Cable Tray						

SPECIFICATIONS FOR BULK ITEMS (INSTRUMENTATION)

SL NO	ITEM SHORT DESCRIPTION	SPECIFICATION	BBOU	ELEP	AKAB	Total	UNITS
8.1	Cable Tray 50 mm (W) x 25 mm (H) x 2.5 mm thick , SS316	Cable Tray Perforated 50 mm (W) X 25 mm (H) x 2.5 mm thick; MOC-SS 316 ; Std Length 3mtr	200	250	100	400	mtrs
8.2	Cable Tray 100 mm (W) x 50 mm (H) x 2.5 mm thick SS 316	Cable Tray Perforated 100 mm (W) X 50 mm (H) x 2.5 mm thick; MOC-SS 316 ; Std Length 3mtr	150	150	300	750	mtrs
9	Instrument Hook Up and Instalation Material						
9.1	Double Block and Bleed Monoflange	DBB Mono Flange, 2", 900# RTJ, SS316	15	20	10	45	Nos.
9.2	¼" OD X ¼" NPT Male Connector	Male Connector ¼" OD X ¼" NPT, SS316	15	20	10	45	Nos.
9.3	¼" OD Tube 0.065" THK	Tube . 1/4" OD, 0.065" THK, SS316	8	10	6	24	Meters
9.4	¼" OD Cap	Cap 1/4" OD,SS316	15	20	10	45	Nos.
9.5	Base Plate	Base Plate : 350mm x 350mm x 10mm THK, MS	16	20	6	42	Nos.
9.6	Mounting Plate	Mounting Plate : 200 MM X 125 MM X 3 MM, MS	14	18	4	36	Nos.
9.7	Rib Plate	Rib Plate : 75mm x 200mm x 6mm THK, MS	26	30	10	66	Nos.
9.8	Pipe	Pipe : 2" NB, MS	24	26	10	60	Meters
9.9	Top Plate	Top Plate Ø63 X 3MM : MS	12	20	8	40	Nos.
9.10	U-Bolt,Nuts & Washers	U Bolt : M8 , SS316	30	40	20	90	Nos.
9.11	Bolt, Nuts, Washers & Spring Washers	Bolt and Nut : M16x110 LONG, SS316	30	40	20	90	Nos.
9.12	Bolt, Nuts, Washers & Spring Washers	Bolt and Nut : M6, SS316	30	40	20	90	Nos.
9.13	1.5M LG MS CHANNEL	Channel : 100mm x 50mm x 6mm THK, MS	8	10	4	22	Nos.
9.14	1100mm LG ANGLE IRON	50mm x 50mm x 6mm THK	8	10	4	22	Nos.
9.15	Steel Plate	Steel Plate : 350mm x 430 0mm x 6 mm THK, MS	8	10	4	22	Nos.